A fully monolithic formulation for multiphase fluid-structure interaction

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Computational modeling of fluid-structure interaction as well as multiphase fluid flow represent highly relevant approaches for insights into many engineering systems, but few works have thoroughly studied the combined effects from a numerical perspective. To-date approaches are either partitioned schemes [3] or fully Eulerian [2], limiting solver robustness or resolution of the fluid-solid interface. We present a first unified and monolithic finite element approach to multiphase fluid-structure interaction, where the flow is described by a coupled five-field Cahn-Hilliard Navier-Stokes equation system [1] in Arbitrary Lagrangian-Eulerian (ALE) description, and the structure is governed by finite strain elastodynamics. Unknowns of the resulting six-field system are fluid velocities, pressures, phase field, chemical potential, domain displacements, and structural deformation. Coupling of fluid and solid is achieved with a monolithic Neumann-Dirichlet scheme, where the structure is constrained by fluid kinematics and the fluid receives the reaction forces, circumventing the introduction of a Lagrange multiplier. Avenues for the effective solution of the resulting system are shown, and applications to elasto-capillarity and bubble-membrane interactions in electrolyzers are demonstrated.

References

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